

10.29 Given $f(x) = \ln(x)$, the first derivative is:

$$\frac{df(x)}{dx} = \frac{1}{x}$$

and the second derivative is:

$$\frac{d^2 f(x)}{dx^2} = -\frac{1}{x^2}$$

For $0 < x < \infty$, the second derivative will always be negative, making it concave.

We find the dual using (10.133):

$$g(\lambda) = \min_x \{\lambda x - f(x)\}$$

Taking the derivative w.r.t x ,

$$\begin{aligned} &= \frac{d}{dx} \{\lambda x - \ln x\} \\ &= \{\lambda - 1/x\} \end{aligned}$$

Setting it to 0, we get : $x = \frac{1}{\lambda}$

$$\begin{aligned} \Rightarrow g(\lambda) &= \left\{ \lambda \frac{1}{\lambda} - \ln \left(\frac{1}{\lambda} \right) \right\} \\ &= 1 + \ln \lambda \end{aligned}$$

Minimizing $\lambda x - g(\lambda)$ w.r.t λ , similar to what we did above, we use (10.132) here:

$$f(x) = \min_{\lambda} \{\lambda x - g(\lambda)\}$$

Taking the derivative w.r.t λ ,

$$\begin{aligned}
&= \frac{d}{d\lambda}(\lambda x - g(\lambda)) \\
&= x - g'(\lambda) \\
&= x - \frac{d}{d\lambda}(1 + \ln \lambda) \\
&= x - \frac{1}{\lambda}
\end{aligned}$$

Setting it to 0, we get : $\lambda = \frac{1}{x}$

$$\begin{aligned}
\Rightarrow f(x) &= \left\{ \frac{1}{x}x - \left(1 + \ln \left(\frac{1}{x} \right) \right) \right\} \\
&= \{1 - (1 - \ln x)\} \\
&= \ln x
\end{aligned}$$

thus recovering the original function.